Dr. M. RAMESH
Associate Professor

Contact

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Academic Qualifications

Degree	University	Year	Subjects	Percentage
B.Sc.,	The Madura College (Autonomous), Madurai – 625 001.	1987	Botany	I
M.Sc.,	School of Biological Sciences Madurai Kamaraj University, Madurai – 625 021.	1990	Biology	I
M.Phil.,	School of Energy Sciences Madurai Kamaraj University, Madurai – 625 021.	1991	Energy, Environment & Natural Sciences	I

Ph.D.,	Centre for Plant Molecular Biology (CPMB), School of Biotechnology, Madurai Kamaraj University, Madurai – 625 021.	1998	Biotechnology	Highly Commended

Teaching Experience

Designation & Office Address	Period	
	From	**To**
Associate Professor Department of Biotechnology, Alagappa University, Karaikudi 630 003.	12.05.2015	Till Date
Assistant Professor (Stage III) Department of Biotechnology, Alagappa University, Karaikudi 630 003.	12.05.2012	11.05.2015
Assistant Professor (Stage II) Department of Biotechnology, Alagappa University, Karaikudi 630 003.	12.05.2007	11.05.2011
Lecturer Department of Biotechnology Alagappa University, Karaikudi 630 003.	12.05.2003	11.05.2007
Lecturer Department of Biotechnology, Dr. G. R. Damodaran College of Science, Coimbatore – 641 014.	01.06.1999	10.05.2003
Lecturer Department of Microbiology, K.S.R. College of Arts & Science,Tiruchengode - 637 209.	09.03.1998	06.04.1999

Additional Academic & Administrative Responsibilities

1. Member of the Standing Committee on Academic Affairs (SCAA) under Section 24(2) (a) Clause II of the Alagappa University Act 1985, with effect from 30.06.2018.

2. HOD In-charge –Department of Botany, Alagappa University (Since 22.11.2019).

3. Department level SWAYAM & NAD Coordinator for M.Sc. Biotechnology Students and Students of Govt. Arts College, Tiruvadanai (June 2017 to November 2019).

4. Coordinator, Centre for Youth Welfare and Empowerment, Alagappa University (September 2018 – September, 2019).

Research Experience

22 Years

Areas of Research

1. **Genetic modification of food crops (rice and millets) for abiotic stress tolerance and idenification of novel unique and combined abiotic stress (CAbS) responsive genes from crop species.**

2. ***Ex situ* conservation of endangered, threatened, and vulnerable medicinal plants**

3. **Enhancement of commercially important phytochemicals through biotechnological approaches such as hairy root culture and elicitation.**

Research Supervision / Guidance

Program of Study		Completed	Ongoing
Research	Ph.D.	11	4
	M.Phil.	3	Nil
Project	PG	59	Nil
	UG / Others (ADMD)	1	Nil

Publications

International		National		Others
Journals	Conferences	Journals	Conferences	Books / Chapters / Monographs / Manuals
54	35	10	98	15

Cumulative impact factor : **121.798**

h-index : **18**

i10 index : **29**

Total Citations : **858 (as on 01.06.2020)**

Thomson Reuters Researcher ID : **P-9021-2017**

Scopus Author ID : **8543374000**

URL:http://www.researcherid.com/rid/P-9021-2017

ORCHID: http://orcid.org/000-0002-7969-4935

Funded Research Projects

Completed Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1.	SERB (Science and Engineering Research Board)	11.07.2013	10.07.2016	Micropropagation of Nilgirianthus *ciliatus* (Nees) Bremek	15.28
2.	CSIR (Councilof Scientific & Industrial Research, New Delhi)	01.05.2012	31.04.2015	Development of abiotic stress tolerant *indica* rice lines (IR 64) with Stress Associated Protein gene (SAP8) through *Agrobacterium* – mediated transformation.	24.52
3.	UGC (University Grants Commission, New Delhi)	01.02.2009	31.01.2012	*In vitro* selection and *Agrobacterium* - mediated transformation studies for developing drought and salinity tolerant *indica* rice (*Oryza sativa* L.)	12.30
4.	UGC (University Grants Commission, New Delhi)	01.04.2007	31.03.2008	*In vitro* micropropagation from different explants of *Jatropha curcas*. L – A potential energy source for the mass production of renewable energy (Bio-diesel).	0.95
5.	DBT (Department of Biotechnology, Special Programme Division,New Delhi)	01.04.2001	31.03.2003	Cultivation of Oyster Mushroom & Processing	9.38

Ongoing Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1.	Board of Research in Nuclear Sciences (BRNS), Department of Atomic Energy, Mumbai.	2020	2023	Radiation induced enhancement of anticancer triterpenoids of *Nilgirianthus ciliatus*	28.79
2.	Rashtriya Uchchatar Shiksha Abhiyan (RUSA) Centrally Sponsored Scheme. RUSA 2: Theme based research project (4.4.2) in Biological Sciences for Translational Health Research for Human, Animal & Plant Systems (4.4.2.1.2.4)	28.01. 2019	27.01.2021	Enhanced and sustainable production of high value brain boosting phytochemicals of *Bacopa monnieri* (L.) Pennell through Hairy Root and Elicitation Technology	10.94

Distinctive Achievements / Awards

1. Ananthalakshmi Ramulu Iyer Prize for Proficiency in Botany (B.Sc.,) – 1988.
2. Narasimha Iyengar Prize for Proficiency in Chemistry and Zoology (B.Sc.,) – 1988.
3. National Eligibility Test (NET) for Junior Research Fellowship & Lectureship conducted by Council of Scientific & Industrial Research(CSIR) and University Grants Commission(UGC), Government of India, New Delhi (Register No.12568; Ref No.2-10/92- E.U.11/ Dec, 91 dt.05.05.1992).
4. Eminent Scientist Award by Biologix Research & Innovation Centre Pvt. Ltd (BRICPL) India – 2017.
5. Bose Science Society Award, Tamilnadu Scientific Research Organization, Pudukottai – 2017.
6. Fellow of Bose Scientific Society (FBBS)) F 300/BSS/2017)dt. 11,01.2017.
7. Alagappa Excellence Award for Research (2018) (Certificate, Citation and Gold Medal and Cash award of Rs. 15000) – 2018.

Events organized in leading roles

Number of Seminars / Conferences / Workshops / Events organized: **2**

Membership in Professional Bodies

1. Life Member: Society of Biological Chemists, India (1740).
2. Life Member: The Indian Science Congress Association, India (L.14559).
3. Life Member: Proteomics Society, India (Reg No.127).
4. Life Member: Biotech Research Society, India (LM 1667).
5. Life Member: Bose Science Society, Tamilnadu Scientific Research Organization (M 300/BSS/ 2017).
6. Life Member: The Indian Botanical Society (135).
7. Life Member: National Academy of Biological Sciences (NABS), India (LM 045-18).
8. Member : International Natural Product Science Taskforce (INPST).

Advisory &Editorial Board

1. Academic Editor - International Journal of Plant & Soil Science (ISSN: 2320-7035).
2. Advisory Board Member - Medicinal Plants – International Journal of Phytomedicines & Related Industries (ISSN: 0975-4261).
3. Editorial Board Member - Science Domain International - An Open Peer Reviewed Journal in Science, Technology & Medicine (www.sciencedomain.org).
4. Editorial Committee Member (ECM) - Indian Journal of Natural Sciences (IJNS) ISSN: 0976-0997, NAAS rated score: 3.56, MJL Thomson Reuters, USA (Since 23, January 2017 onwards).
5. Advisory Board Member of Namadhu Ariviyal, a Science Awareness Tamil Monthly.

Academic Bodies (such as Boared of Studies etc.,)

1. Member of the Board of Studies in B.Sc., Biotechnology, Alagappa University, since May 2012.
2. Member of the Board of Studies in M.Sc., Biotechnology , Alagappa University, Since March, 2003.
3. Member of the Board of Studies in M.Phil., Biotechnology, Alagappa University, Since July, 2013.
4. Member of the Board of Studies in M.Sc., Botany of Affiliated Colleges of Alagappa University, Since December, 2011.
5. Member of the Board of Studies in B.Sc., Biochemistry, Alagappa University, Since May, 2011.
6. Member of the Board of Studies in B.Sc., Zoology (Animal Biotechnology) of Affiliated Colleges of Alagappa University, Since February, 2009.
7. Member of the Board of Studies in B.Sc., and M.Sc., Biotechnology, SNR College, Coimbatore, Since June 2013.
8. Deputy Coordinator of Intellectual Property Rights Cell, Alagappa University (2013 – 2016).
9. Member of the Board of Studies in U.G. and P.G. Biotechnology of J.J. College of Arts and Science, Pudukkottai, Since February, 2019.
10. Examination Squad member- AU Colleges, November 11, 2019.

Expert / Inspection /Assessment Coimmittee Member:

1. Village Placement Program Coordinator (2005, 2008, 2012, 2014).
2. Subject Expert for the interview committee for recruiting Lecturer in Botany at P.T.M.T.M. College, Kamuthi (2009).
3. Assessment Committee Member to assess the work of A. Kalaiarasi for the upgradation of JRF to SRF under the Rajiv Gandhi National Fellowship scheme held on 15.10.2009.
4. Expert Committee Member to assess the academic and infrastructural facilities for recommending affiliation of M.Sc., Botany Programme to Department of Botany, Alagappa Govt. Arts College, Karaikudi (2011).
5. Inspection Committee member to assess the infrastructure and expertise available at the Department of Botany, Alagappa Arts College, Karaikudi for recognition as Approved Research Centre of Alagappa University (2011).
6. Expert Committee Member to assess the Course contents of Botany, Samacheer Kalvi Subjects Workshop held at Alagappa Matriculation Higher Secondary School, Karaikudi (2011).
7. Inspection Committee member to assess the infrastructure and expertise available at the Department of Botany, Pasumpon Thiru Muthuramalinga Thevar Memorial College, Kamuthi (2012).
8. DBT outside External Expert in IBSC (Institutional Biosafety Committee) for DST Project at Ayya Nadar Janaki Ammal College (2013).
9. Inspection Committee Member for recognition of Approved Research Centre for Botany in Affiliated Colleges of Alagappa University (2013).
10. Member of the Inspection Squad for Alagappa University Examinations of Affiliated Colleges of Alagappa University (2015).
11. Inspection Committee member to inspect the academic and infrastructural facilities for the grant of affiliation of B.Sc. Biotechnology Programme, Vidhyaa Giri College of Arts and Science, Puduvayal (2017).
12. Acted as reviewer to assess DBT (GoWB - West Bengal Govt) proposal entitled "DNA Fingerprint of fresh leaves of medicinal plants by random amplified polymorphic DNA (RAPD) – PCR Method for Authentication" (2017).
13. Member of Doctoral Committee for the Biotechnology Ph.D., Student Mr. Rajinikanth Garapati (Guide : Dr. M.Ramesh) of JJ College Pudukkottai (2012).
14. Member of Doctoral Committee for the Microbial Biotechnology Ph.D., Student Ms. B. Thazeem (Guide : Dr K. Preethi) of Bharathiar University, Coimbatore (2016).
15. Doctoral committee External Expert member for the Biotechnology student Mr. M. Saravanan (Giude: Dr.R. Sathishkumar) of Bharathiar University, Coimbatore (2017).
16. Member of the Advisory Committee and Plenay Speaker at the International Conference on Innovative and Emerging Trends in Botant (ICIETB 2019) organized by the Dept. of Botany, Alagappa University (6 - 7, November , 2019).
17. Examination Squad member- AU Colleges (November 11, 2019).

18. TANSCHE Expert Committee member to give equivalence to B.Sc. and M.Sc. Plant Biology and Biotechnology courses offered by Bharathidasan and Bharathiar University (3.12.2019).
19. DRC Member to Mr.Raja (Guide: Dr. Hema Jagadesan), PSG College of Technology, Coimbatore (24.1.2020).
20. Inspection Committee member to recognize N. Ramavarier Ayurveda Foundation, Madurai for recognizing the Department of Biotechnology as a Research Centre of Alagappa University (12.02.2020).

Reviewer:

Reviewer for Journals such as Frontiers In Plant Science, PLOS ONE, Applied Biochemistry and Biotechnology (Springer, US), Plant Cell Tissue and Organ Culture (Kluwer academic Publishers, Netherlands), Journal of Environmental Biology (Lucknow, India), Journal of Biotechnology (Elsevier, Germany), Acta Physiologiae Plantarum (Springer, Poland), Trees (Springer, Germany), Indian Journal of Experimental Biology (NISCAIR, CSIR, India), Rice Science (Elsevier, China), Journal of Medicinal Plants Research (Academic Journals), Natural Product Radiance (CSIR, NISCAIR, India), Maejo International Journal of Science and Technology (Japan), Agricultural Science Research (Saudi Arabia), Physiology and Molecular Biology of Plants (Springer, India), Journal of Applied Phycology (Springer, Netherlands), Rice Science, Gene Reports, Journal of Biotechnology and Molecular Biology, 3Biotech (Springer India), Flavour and Fragrance, SpringerPlus, Plant Cell Biotechnology and Molecular Biology, Pharmaceutical Biology (Taylor & Francis), Physiology and Molecular Biology of Plants, BMC Genomics, Plant Cell Biotechnology and Molecular Biology, Genome, Horticultural science,Biologia, Planta, The Journalof Horticultural Science and Biotechnology, *In vitro* Cell & Developmental Biology- Plant.

Others

1. Number of Invited / Special Lectures delivered: 34

2. Number of Seminars / Conferences / Workshops organized:2

3. No. of PhD Thesis evaluated : 12

4. No. of PhD Public Viva Voce Examination conducted : 15

5. Novel unique, combined abiotic stress (CAbS) and combined abiotic stress transcription factor genes identified: 344

6. Popular article in Regional Languages: Published an article entitled "Biotechnology Challenges and Achievements" in Namadhu Ariviyal, a Science Awareness Tamil Monthly.vol:1, Issue :1, March 2019, pp: 8-10.

7. Abstracts Published in Journals: **2**.
   I. Muthuramalingam P**,** Pandian SK and **Ramesh M**. (2017). Global proteomic perspective and identification of abiotic stress responsive proteins in rice: An *in silico* approaches. **Journal of Proteins and Proteomics.** 8, ISSN: 0975 – 8151**. (**Impact Factor – **1.0;** NAAS – **4.55).**
   II. Muthuramalingam P, Jeyasri R, Pandian SK and **Ramesh M**. (2018). Regulatory network analysis of rice MYB transcription factor family proteins: A systems biology perspective. **Journal of Proteins and Proteomics.** 9, ISSN: 0975 – 8151**. (**Impact Factor – **1.0;** NAAS – **4.55).**

8. Proceedings: 1

   Rameshkumar R, Satish L, Rathinapriya P, Sasireka S, Dharmarajan S, **Ramesh M** (2017) Lab to land conservation of IUCN – red listed, vulnerable Indian medicinal plant (*Nilgirianthus ciliatus*) from possible extinction In**:** Proceedings of the International Conference on Herbal and Natural Components as the Future of Pharmacology and the Annual Meet of the National Society of Ethnopharmacology pp. 488 – 493 (ISBN: 978-93-84234-04-1).

Research & Review Articles

1. Rathinapriya P, Pandian S, Satish L, Rameshkumar R, Rakkammal K, **Ramesh M** (2020) Effects of liquid seaweed extracts in improving the agronomic performance of foxtail millet. **Journal of Plant Nutrition** (Taylor & Francis,USA) (**ISSN: 0190-4167)** (Impact Factor: 0.753) (In Press).

2. Velsankar K; Preethi R; Jeevan Ram P S; **Ramesh M** & Sudhahar Sakkarapani (May 2020) Evaluations of Biosynthesized Ag nanoparticles via Allium Sativum flower extract in biological applications . **Applied Nanoscience** (Springer, Switzerland) (Impact Factor: 3.198) (In Press).

3. Muthuramalingam P, Jeyasri R, Kalaiyarasi D, Krishnan SR, Aruni W, Karutha Pandian & **Ramesh M** (10 April, 2020) Global transcriptome analysis of novel stress associated protein (SAP) genes expression dynamism of combined abiotic stresses in *Oryza sativa* (L.). **Journal of Biomolecular Structure & Dynamics** (Taylor & Francis) (doi:10.1080/07391102.2020.1747548 (Impact Factor: 3.310).

4. Jeyasri R, Muthuramalingam P, Suba V, Chen T-J, **Ramesh M** (02, April 2020) Bacopa monnieri and Their Bioactive Compounds Inferred Multi-Target Treatment Strategy for Neurological

Diseases: A Cheminformatics and System Pharmacology Approach, **Biomolecules** (MDPI), 10, 536; doi:10.3390/biom10040536 (Impact Factor: 4. 694).

5. Saha P S, Sarkar S, Jeyasri R, Muthramalingm P, **Ramesh M**, Sumita Jha (26 March, 2020) In vitro propagation phytochemical and neuropharmacological profiles of Bacopa monnieri (L.) Wettest. : A Review , **Plants** (MDPI) , 9, 411; doi:10.3390/plants9040411(Impact Factor: 2.632).

6. Pandian S, Satish L, Shilpha J & **Ramesh M** (05 April, 2020) Genetic diversity analysis reveals strong population structure in Sorghum germplasm collection. **Proceedings of the National Academy of Sciences, India Section B: Biological Sciences**., India, Sect.B Biol.Sci, Jan – March 2020, 90(1) 179 – 190 (doi:10.1007/s 40011-019-01095-9 (mpact Factor :0.396).

7. Muthuramalingam P, Jeyasri R, Bharathi RKAS, Suba V, Pandian STK and **Ramesh M** (06 June 2019). Global integrated omics expression analyses of abiotic stress signaling HSF transcription factor genes in Oryza sativa L.: An in silico approach. **Genomics** (Elsevier), 112(1) 908 – 918 (doi:10.1016/j.ygeno.2019.06.006) (Impact Factor: 3.327).

8. Krishnan SR, Pandian S, Banupriya R, Muthuramalingam P, Banu SJ, Manikandan A & **Ramesh M** (2019) Augmenting competent *in vitro* organogenesis etiquette from leaf base of country mallow, *Abutilon indicum* L. sweet: an ethno-botanically valuable medicinal plant. **Biocatalysis and Agricultural Biotechnology** (Elsevier, Netherlands) April 2019, 19:101125. (Impact Factor: 2.14).

9. Rameshkumar R, Pandian S, Rathinapriya P, Tamil Selvi C, Satish L, Gowrishankar S, Leung D.W.M. & **Ramesh M** (2019) Genetic diversity and phylogenetic relationship of *Nilgirianthus ciliatus* populations using ISSR and RAPD markers: Implications for conservation of an endemic and vulnerable medicinal plant. **Biocatalysis and Agricultural Biotechnology** (Elsevier, Netherlands) March 2019, 18 – 101072 (Impact Factor: 2.140).

10. Rameshkumar R, Karthikeyan A, Rathinapriya P & **Ramesh M** (2019) Micropropagation of traditional deep water rice (*Oryza sativa* L.) cv. TNR1 for viable seed production and germplasm conservation. **Biocatalysis and Agricultural Biotechnology** [Elsevier, Netherlands] 18: 100999 (doi: 10.1016/j.bcab.2019.01.037 (Impact Factor :2.14).

11. Pandian S & **Ramesh M** (2019) Decoding of Finger Millet Genome: A Milestone of Millet Genomics. **Signal Transduction Insights**, (SAGE Publications, USA) 8: 1–3 (doi: 10.1177/1178643418820541).

12. Pandian S, Marichelvam K, Satish L, Ceasar SA, Pandian SK, **Ramesh M** (2018) SPAR markers assisted assessment of genetic diversity and population structure in finger millet (Eleusine coracana (L.) Gaertn) mini core collection. **Journal of Crop Science and Biotechnology** (Springer, Switzerland). December 2018, 21:469-481(doi:10.1007/s12892-018-0034-0).

13. Rathinapriya P, Satish L, Rameshkumar R, Pandian S, Rency SA, **Ramesh M** (2018). Efficient plant regeneration from leaf base segments of foxtail millet (*Setaria italica* (L.) Beauv.) genotypes using activated charcoal and amino acids. **Physiology and Molecular Biology of**

Plants, 25(2):533–548 (March–April 2019) **(**doi:10.1007/s12298-018-0619-z) (Springer, India) (Impact Factor: 1.539).

14. Rameshkumar R, Satish L, Pandian S, Rathinapriya P, Rency AS, Gowrishankar S, Pandian SK, David W. M. Leung, **Ramesh M** (2018). Production of squalene with promising antioxidant properties in callus cultures of *Nilgirianthus ciliatus.* **Industrial Crops and Products** [Elsevier, Ireland] October 2018, 126: 357–367 (doi: 10.1016/j.indcrop.2018.10.031) (Impact Factor: 4.191).

15. Rency AS, Pandian S and **Ramesh M** (2018). Influence of adenine sulphate on multiple shoot induction in *Clitoria ternatea* L. and analysis of phyto-compounds in *in vitro* grown plants. **Biocatalysis and Agricultural Biotechnology** [Elsevier]July 2018, 16: 181-191. (doi: 10.1016/j.bcab.2018.07.034) (Impact Factor: 2.14).

16. Muthuramalingam P, Krishnan SR, Pandian S, Mareeswaran N, Aruni W, Pandian SK and **Ramesh M** (2018). Global analysis of threonine metabolism genes unravel key players in rice to improve the abiotic stress tolerance. **Scientific Reports** [Nature Publishing Group, UK] June 2018, 8: 9270. (doi:10.1038/s41598-018-27703-8) (Impact Factor: 4.259).

17. Pandian S, Satish L, Rameshkumar R, Muthuramalingam P, Rency SA, Rathinapriya P & **Ramesh M** (2018). Analysis of population structure and genetic diversity in an exotic germplasm collection of *Eleusine coracana*(L.) Gaertn. using genic-SSR markers. **Gene** (Elsevier, USA).653: 80 -90. (doi: 10.1016/j.gene.2018.02.018) (Impact Factor:2.638).

18. Krishnan SR, Muthuramalingam P, Pandian S, Banupriya R, Chithra G & **Ramesh M** (2018). Sprouted sorghum extract elicits coleoptile in indica rice and enhances its shoot and root acclimatization, maintaining its genetic fidelity (R-ISSR). **Rice Science** [China National Rice Research Institute, Hangzhou, Elsevier] 25(2):61-72 (doi: 10.1016/j.rsci.2017.08.005) 24(6) ISSN: 1672-6308 (Impact Factor: 2.370).

19. Muthuramalingam P, Krishnan S R, Saravanan K, Mareeswaran N, Kumar R & **Ramesh M** (2018). Genome-wide identification of major transcription factor superfamilies in rice identifies key candidates involved in abiotic stress dynamism. **Journal of Plant Biochemistry and Biotechnology** [Springer, India] (doi:10.1007/s 13562-0440 -3) (Impact Factor: 1.327).

20. Satish L, Rency AS & **Ramesh M** (2018). Spermidine sprays alleviate the water deficit-induced oxidative stress in finger millet (*Eleusine coracana* L. Gaertn.) plants.**3 Biotech** [Springer, India] 8: 1- 11 (doi: 10.1007/s13205-018-1097-2) (Impact Factor: 1.786).

21. Rameshkumar R, Rathinapriya P, Satish L, Pandian S, Rency AS & **Ramesh M** (2017). *In vitro* propagation and conservation of useful endangered medicinal plants with anticancer activity. **Journal of Molecular Biology and Biotechnology** [iMedPub Journals, USA] 2(3):8.

22. Satish L, Santhakumari S, Gowrishankar S, Pandian SK, Ravi AV and **Ramesh M** (2017). Rapid biosynthesized AgNPs from *Gelidiella acerosa* aqueous extract mitigates quorum sensing mediated biofilm formation of Vibrio species - An in vitro and in vivo approach. **Environmental Science and Pollution Research** [Springer, Berlin Heidelberg] (doi: 10.1007/s11356-017-0296-4). (Impact Factor : 2.914).

23. Pandian S, Sivasankar C, Muthuramalingam P & **Ramesh M** (2017) An Amazing nutritional Value in Wonderful Finger Millet Makes This "The Most Lovable Food Crop" to the World, **Scientific Journal of Food Science and Nutrition**, 3(2) 34-36.

24. Pandian S, Muthuramalingam P & **Ramesh M** (2017). ESTs as a resource for gene discovery and population genetic analysis of crop plants. **MOJ Cell Science and Report**. 4 (4): 00096 (doi: 10.15406/mojcsr.2017.04.00096).

25. Muthuramalingam P, Krishnan SR, Pandian S & **Ramesh M** (2017). Emerging trends on abiotic stress tolerance investigation in crop plants. **Advances in Biotechnology and Microbiology.** 6: 1. 6(1): 555678. (doi: 10.19080/AIBM.2017.06.555678) (Impact Factor: 0.734).

26. Satish L, Ceasar SA and **Ramesh M** (2017). Improved *Agrobacterium*-mediated transformation and rapid regeneration in four cultivars of finger millet (*Eleusine coracana* (L.) Gaertn.). **Plant Cell Tissue and Organ Culture** [Springer, Netherlands] 131:547–565. (doi: 10.1007/s11240-017-1305-5) (Impact Factor : 2.28).

27. Muthuramalingam P, Krishnan SR, Pothiraj R & **Ramesh M** (2017). Global transcriptome analysis of combined abiotic stress signaling genes unravels key players in *Oryza sativa* L.: An *in silico* approach. **Frontiers in Plant Science.** 8:759. (doi: 10.3389/fpls.2017.00759) (Impact Factor :4.495).

28. Rency AS, Satish L, Pandian S, Rathinapriya P & **Ramesh M** (2017). *In vitro* propagation and genetic fidelity analysis of alginate encapsulated *Bacopa monnieri* shoot tip using *Gracilaria salicornia* extracts. **Journal of Applied Phycology** [Springer, Netherlands] 29:481- 494 (doi: 10.1007/s10811-016-0918-0) (Impact Factor: 2.635).

29. Rameshkumar R, Largia M V, Satish L, Shilpha J & **Ramesh M** (2017). *In vitro* mass propagation and conservation of *Nilgirianthus ciliatus* through nodal explants: A globally endangered, high trade medicinal plant of Western Ghats. **Plant Biosystems** [Taylor & Francis, United Kingdom] 151 (2), 204- 211 (doi: 10.1080/11263504.2016.1149120) (Impact Factor: 1.525).

30. Largia MJV, Satish L, Johnsi R, Shilpha J & **Ramesh M** (2016). Analysis of propagation of *Bacopa monnieri* (L.) from hairy roots, elicitation and Bacoside A contents of Ri transformed plants. **World Journal of Microbiology and Biotechnology** [Springer, Netherlands] 32:1-11 (doi:10.1007/s11274-016-2083-7) (Impact Factor: 2.652).

31. Sivaranjani M, Krishnan S R, Kannappan A , Ramesh M, Veera Ravi A (2016) Curcumin from Curcuma longa affects the virulence of Pectobacterium wasabiae and P. carotovorum subsp. carotovorum via quorum sensing regulation. **European J of Plant Pathology** [Koninklijke Nederlandse Planteziektenkundige Vereniging] 146:793–806 (doi: 10.1007/s10658-016-0957-z) (Impact Factor: 1.744).

32. Satish L, Shilpha J, Pandian S, Rency SA, Rathinapriya P, Ceasar SA, Largia MJV, Rameshkumar R, Kumar AA & **Ramesh M** (2016). Analysis of genetic variation in Sorghum (*Sorghum bicolor*

L. Moench) genotypes with various agronomical traits using SPAR methods. **Gene** [Elsevier, Ireland], 576,581-585. (doi: 10.1016/j.gene.2015.10.056) (Impact Factor: 2.638).

33. Satish L, Rency AS, Rathinapriya P, Ceasar SA, Pandian S, Rameshkumar R, Rao TB, Balachandran SM and **Ramesh M** (2016). Influence of plant growth regulators and spermidine on somatic embryogenesis and plant regeneration in four Indian genotypes of finger millet (*Eleusine coracana* (L.) Gaertn). **Plant Cell Tissue and Organ Culture** [Springer,Netherlands]124:15–31(doi: 10.1007/s11240-015-0870-8) (Impact Factor: 2.390).

34. Satish L, Rathinapriya P, Rency SA, Ceasar SA, Prathibha M, Pandian S, Rameshkumar R and **Ramesh M** (2016). Effect of salinity stress on finger millet (*Eleusine coracana* (L) Gaertn): histochemical and morphological analysis of coleoptile and coleorhizae. **Flora – Morphology, Distribution, Functional Ecology of Plants** [Elsevier, Germany] 222:111- 120 (doi: 10.1016/j.flora.2016.04.006) (Impact Factor: 1.590).

35. Shilpha J, Jayashre M, Largia MV and **Ramesh M** (2016). Direct shoot organogenesis and *Agrobacterium tumefaciens* mediated transformation of *Solanum trilobatum* L. **Turkish journal of Biology** [TUBITAK Academic Journals] 40: 866-877 (doi: 10.3906/biy-1509-83) (Impact Factor: 1.183).

36. Satish L, Rathinapriya P, Rency AS, Ceasar SA, Pandian S, Rameshkumar R and **Ramesh M** (2016). Somatic embryogenesis and regeneration using *Gracilaria edulis* and *Padina boergesenii* seaweed liquid extracts and genetic fidelity in finger millet (*Eleusine coracana*). **Journal of Applied Phycology** [Springer, Netherlands] 28:2083- 2098 (doi: 10.1007/s10811-015-0696-0) (Impact Factor**:** 2.635).

37. Satish L, Rathinapriya P, Ceasar SA, Rency AS, Pandian S, Rameshkumar R, Subramanian A and **Ramesh M** (2016). Effects of cefotaxime, amino acids and carbon source on somatic embryogenesis and plant regeneration in four Indian genotypes of foxtail millet (*Setaria italica* L.). **In Vitro Cellular & Developmental Biology - Plant** [Kluwer academic Publishers, USA] 52,140 - 153 (doi: 10.1007/s11627-015-9724-7) (Impact Factor: 1.454).

38. Shilpha J, Satish L, Kavikkuil M, Largia MJV and **Ramesh M** (2015). Methyl jasmonate elicits the solasodine production and anti-oxidant activity in hairy root cultures of *Solanum trilobatum* L. **Industrial Crops and Products** [Elsevier, Ireland] 71:54–64 (doi: 10.1016/j.indcrop.2015.03.083) (Impact Factor: 4.191).

39. Largia MJV, Pothiraj G, Shilpha J and **Ramesh M** (2015). Methyl Jasmonate and Salicylic acid synergism enhances Bacoside A content in shoot cultures of *Bacopa monnieri* (L.). **Plant Cell Tissue and Organ Culture** [Kluwer academic Publishers, Netherlands] 122:9-20 (doi: 10.1007/s11240-015-0745-z) (Impact Factor: 2.69).

40. Largia MJV, Shilpha J, Pothiraj G and **Ramesh M** (2015). Analysis of nuclear DNA content, genetic stability, Bacoside A quantity and antioxidant potential of long term *in vitro* grown germplasm lines of *Bacopa monnieri* (L.). **Plant Cell Tissue and Organ Culture** [Kluwer academic Publishers, Netherlands] 120:399–406 (doi: 10.1007/s11240-014-0602-5) (Impact Factor: 3.633).

41. Priya AM Krishnan SR and **Ramesh M** (2015). Ploidy stability of *Oryza sativa* L. cv IR64 transformed with moth bean *P5CS* gene with significant tolerance against drought and salinity. **Turkish Journal of Biology** [Scientific & Technological Research Council of Turkey] 39:407- 416 (doi: 10.3906/biy-1409-43) (Impact Factor: 1.216).

42. Satish L, Ceasar SA, Shilpha J, Rency AS, Rathinapriya P and **Ramesh M** (2015). Direct plant regeneration from *in vitro* - derived shoot apical meristems of finger millet (*Eleusine coracana* (L.) Gaertn. **In Vitro Cellular & Developmental Biology - Plant** [Springer, US] 51:192–200 (doi: 10.1007/s11627-015-9672-2) (Impact Factor: 1.454).

43. Satish L, Rameshkumar R, Rathinapriya P, Pandian S, Rency AS, Sunitha T and **Ramesh M** (2015). Effect of seaweed liquid extracts and plant growth regulators on *in vitro* mass propagation of brinjal (*Solanum melongena* L.) through hypocotyl and leaf disc explants. **Journal of Applied Phycology** [Springer, Netherlands] 27:993–1002 (doi: 10.1007/s10811-014-0375-6) (Impact Factor: 2.635).

44. Shilpha J, Silambarasan T, Largia MJV and **Ramesh M** (2014). Improved *in vitro* propagation, solasodine accumulation and assessment of clonal fidelity in regenerants of *Solanum trilobatum* L. by flow cytometry and SPAR methods. **Plant Cell Tissue and Organ Culture** [Kluwer academic Publishers, Netherlands] 117:125–129 (doi: 10.1007/s11240-013-0420-1) (Impact Factor: 3.28).

45. Largia MJV, Pandian SK and **Ramesh M** (2013). Genetic fidelity assessment of encapsulated *in vitro* tissues of *Bacopa monnieri* after 6 months of storage by using ISSR and RAPD markers. **Turkish Journal of Botany** [TUBITAK, Turkey] 37:1008-1017 (doi: 10.3906/bot-1207-24) (Impact Factor: 1.6).

46. Shilpha J, Silambarasan T, Pandian SK and **Ramesh M** (2013). Assessment of genetic diversity in *Solanum trilobatum* L., an important medicinal plant from South India using RAPD and ISSR markers. **Genetic Resources and Crop Evolution** [Springer, Netherlands] 60:807-818 (doi: 10.1007/s10722-012-9951-2) (Impact Factor: 1.593).

47. Krishnan SR, Priya AM and **Ramesh M** (2013). Rapid regeneration and ploidy stability of 'cv IR36' *indica* rice (*Oryza sativa* L.) confers efficient protocol for *in vitro* callus organogenesis and *Agrobacterium tumefaciens* mediated transformation. **Botanical Studies** [Springer open] 54:47 (doi: 10.1186/1999-3110-54-47) (Impact Factor: 1.796).

48. Karthikeyan A, Rameshkumar R, Sivakumar N, Ali Amri IS, Pandian SK and **Ramesh M** (2012). Antibiofilm activity of *Dendrophthoe falcata* against different bacterial pathogens. **Planta Medica** [Georg Thieme Verlag KG Stuttgart. New York] 78:1918-1926 (doi: 10.1055/s-0032-1327879) (Impact Factor: 2.746).

49. Priya AM, Pandian SK and **Ramesh M** (2012). Effect of different antibiotics on the elimination of Agrobacterium and high frequency *Agrobacterium*-medicated transformation of *indica* rice *Oryza sativa* (L.). **Czech J Genetics Plant Breeding** [Ministry of Agriculture of the Czech Republic] 48(3):120–130 (doi: 10.17221/77/2011-CJGPB) (Impact Factor : 0.652).

50. Karthikeyan A, Shilpha J, Karutha Pandian S and **Ramesh M** (2012). *Agrobacterium*-mediated transformation of *indica* rice cv. ADT 43. **Plant Cell Tissue and Organ Culture** [Kluwer academic Publishers, Netherlands] 109:153 – 165 (doi: 10.1007/s11240-011-0083-8) (Impact Factor :2.15).

51. Karthikeyan A, Pandian SK and **Ramesh M** (2011). Transgenic *indica* rice cv. ADT 43 expressing a *Δ1- pyrroline – 5-carboxylate synthetase* (P5CS) gene from *Vigna aconitifolia* demonstrates salt tolerance. **Plant Cell Tissue and Organ Culture** [Kluwer academic Publishers, Netherlands] 107: 383-395 (doi: 10.1007/s11240-011-9989-4) (Impact Factor : 2.15).

52. Karthikeyan A, Pandian SK and **Ramesh M** (2011). *Agrobacterium* –mediated transformation of leaf base derived callus tissues of popular *indica* rice (*Oryza sativa* L.sub sp.*indica* cv.ADT 43). **Plant Science** [Elsevier Ireland] 181: 258 – 268 (doi: 10.1016/j.plantsci.2011.05.011) (Impact Factor: 3.712).

53. Priya AM and **Ramesh M** (2011). Efficient *in vitro* plant regeneration through leaf base derived callus cultures of abiotic stress sensitive popular asian *indica* rice cultivar IR 64 (*Oryza sativa* L.). **Acta Biologica Hungarica** [Akadémiai Kiadó] 62(4) 441 – 452 (doi: 10.1556/ABiol.62.2011.4.9) (Impact Factor : 0.793).

54. Priya AM, Pandian SK and **Ramesh M** (2011). Effect of NaCl on *in vitro* plant regeneration from embryogenic callus cultures of 'cv IR 64' *indica* rice (*Oryza sativa* L.). **African Journal of Biotechnology** [Academic Journals] 10 (36): 6947 – 6953 (Impact Factor: 0.573).

55. Karthikeyan A, Madhanraj G, Pandian SK and **Ramesh M** (2011). Genetic similarity among highly endangered *Bacopa monnieri* (L.) Pennell from Southern India as detected using RAPD analysis. **Genetic Resources and Crop Evolution** [Springer, Netherlands] 58:769 – 782 (doi: 10.1007/s10722-011-9695-4) (Impact Factor: 1.538).

56. **Ramesh M**, Karthikeyan A, Vijayakumar KP, Joe Virgin Largia and Pandian SK (2011). Agrobacterium - mediated transformation of pharmaceutically important Indian medicinal herb *Bacopa monnieri* (L.). **Journal of Medicinal plants Research** [Academic Journals] 5(11): 2316 – 2321 (Impact Factor: 1.539).

57. **Ramesh M**, Vijayakumar KP, Karthikeyan A and Pandian SK (2011). RAPD based genetic stability analysis among micropropagated, synthetic seed derived and hardened plants of *Bacopa monnieri* (L.): a threatened Indian medicinal herb. **Acta Physiologiae Plantarum** [Springer, Poland] 33:163 – 171 (doi: 10.1007/s11738-010-0534-6) (Impact Factor: 1.639).

58. Karthikeyan A, Pandian SK and **Ramesh M** (2009). High frequency plant regeneration from embryogenic callus of a popular *indica* rice (*Oryza sativa* L.). **Physiology and Molecular Biology of Plants** [Springer, India] 15(4) 371 – 375 (doi: 10.1007/s12298-009-0042-6) (Impact Factor: 1.539).

59. **Ramesh M**, Murugiah, and Gupta AK (2009). Efficient plant regeneration from leaf base segments of *indica* rice, **Indian Journal of Experimental Biology** [NISCAIR, CSIR, India] 47: 68 – 74 (Impact Factor: 0. 934).

60. **Ramesh M**, Selvam A and Pandian SK (2007). Cytokinin induced multiple shoot induction from node explants of *Daemia extensa* (Jacq.) R.B – A potentially important medicinal plant. **Asian Journal of Bioscience** 2(1): 18 – 21.

61. **Ramseh M**, Marx R, Mathan G and Pandian SK (2008). Effect of bavistin on *in vitro* plant conversion from encapsulated uninodal microcuttings of micropropagated *Bacopa monnieri* (L.) – An ayurvedic herb. **Journal of Environmental Biology** [Triveni Enterprises, Lucknow (India) 30(3):441-444 (Impact Factor: 1.359).

62. **Ramesh M** and Gupta AK (2006). Genetic transformation of embryogenic callus cultures of *indica* rice using particle bombardment and selection of stably transformed hygromycin resistant plants. **Asian Journal of Microbiology, Biotechnology and Environmental Science** [Scientific Publishers, India; ISSN: 0972-3005] 8(3):617–622.

63. **Ramesh M**, Saravanakumar RM and Pandian SK (2006). Benzyl amino purine and Adenine sulphate induced rapid multiple shoot and root induction from nodal explants of brahmi {*Bacopa monnieri* (L.) Wettst}. **Natural Product Radiance** [NISCAIR, CSIR, India] 5(1):44–51.

64. **Ramesh M** and Gupta AK (2005).Assessment of transient expression of *β - glucuronidase* gene in callus cultures after different stages of cobombardment. **African Journal of Biotechnology** [Academic Journals] 4(7):596–600 (Impact Factor: 0.573).

Chapters in Books:

1 Muthuramalingam P, Jeyasri R, Pandian SK and **Ramesh M** (2010) Bioinformatics and Integrating Omics Approach to Unveil the Regulation of Combined Abiotic Stress in C3 and C4 Cereal Crops. **Cereals.** Springer Nature (In Press).

2 Pandian S ,Rakkammal K, Rency AS, Muthuramalingam P, Pandian SK & **Ramesh M** (2020) Abiotic stress and applications of Omics approaches to develop stress tolerance in Agronomic crops. In **Agronomic Crops,**Vol 3: Stress responses and Tolerance , Editor: M. Hasanuzzaman, ISBN 978-981-15-0024-4, https://doi.org/10.1007/978-981-15-0025-1_26. Springer Nature Singapore Pvt Ltd. 2020. pp:557 – 578.

3 Pandian S and **Ramesh M** (2020) Development of pesticide resistance in pests : A key challenge to the crop protection and environmental safety.In **Pesticides in Crop Protection:Physiological and Biochemical Action.** Editor: Prabhat Kumar Srivastava, ISBN 9781119432197. 2020 John Wiley &Sons Ltd,USA. pp 1-13.

4 Muthuramalingam P, Jeyasri R, Krishnan SR, Pandian SK, Sathishkumar R and **Ramesh M** (November 25, 2019). Integrating Bioinformatics and Omics tools for systems analysis of abiotic stress tolerance in *Oryza sativa* (L.). In: **Advances in Plant Transgenics: Methods and Applications.** ISBN 978-981-13-9623-6, doi.org/10.1007/978-981-13-9624-3 Springer Nature. pp: 59 – 78.

5 Satish L, Rency AS,Rameshkumar R, Swamy MK, **Ramesh M** (October 2019) "Transgenic Plant Cell Cultures: A Promising Approach for Secondary Metabolite Production" In **Natural Bio-active Compounds,** Volume 3, Biotechnology, Bioengineering, and Molecular Approaches, ISBN: 978-981-13-7437- 1, doi: 10.1007/978-981-13-7438-8. Editors: Akhtar**,** Mohd Sayeed**,** Swamy**,** Mallappa Kumara (Eds.) Springer Nature Singapore Pte Ltd. pp: 79 - 122.

6 Rency AS, Pandian S,Rakkammal K, Satish L, Swamy MK and **Ramesh M** (October 2019) "Hairy root cultures as an alternative source for the production of high - value secondary metabolites" in **Natural Bio-active Compounds,** Volume 3, Biotechnology, Bioengineering, and Molecular Approaches, ISBN: 978-981-13-7437- 1, doi: 10.1007/978-981-13-7438-8. Editors: Akhtar**,** Mohd Sayeed**,** Swamy**,** Mallappa Kumara (Eds.) Springer Nature. pp: 237- 264.

7 Muthuramalingam P, Radhesh Krishnan S, Deepak Kumar V and **Ramesh M** (December 18, 2018) Technological Development for Abiotic Stress in Rice: A Critical Overview. In: **Rice Science – Biotechnological and Molecular Advancements**. pp 69-91. ISBN: 978-1-351-13658-7(eBook) Edited by Deepak Kumar Verma, Prem Prakash Srivastav and Nadaf Altafhusain Balechand. Apple Academic Press, USA & CRC Press, a Taylor and Francis Group.pp:69 – 92.

8 Radhesh Krishnan S, Muthuramalingam P, Chakravarthi M and **Ramesh M** (December 18, 2018) Emerging Trends of A20/AN1 Zinc-finger Proteins in Improving Rice Productivity under Abiotic Stress. In: **Rice Science – Biotechnological and Molecular Advancements**. pp 03-27. ISBN: 978-1-351-13658-7(eBook) Edited by Deepak Kumar Verma, Prem Prakash Srivastav and Nadaf Altafhusain Balechand. Apple Apple Academic Press, USA & CRC Press, a Taylor and Francis Group.pp:3 – 28.

9 Radhesh Krishnan S, Muthuramalingam P, Sivamaruthi BS, Chakravarthi M and **Ramesh M** (Sepember 17, 2018). " Genetic Engineering for Fragrance in Rice: An Insight on Its Status" In: **Science and technology of Aroma, Flavor and Fragrance in Rice**. ISBN 13: 978-0-203-71145-3(eBook).. Edited by Deepak Kumar Verma and Prem Prakash Srivastav. Apple Academic Press, USA & CRC Press, a Taylor and Francis Group. pp 295- 320.

10 Muthuramalingam P, Jeyasri R, Kalaiyarasi D, Pandian S, Krishnan SR, Pandian SK and **Ramesh M** (March 28, 2018) Emerging advances in computational omics tools for systems analysis of Gramineae Family grass species and their abiotic stress responsive function. In: **Omics Based Approaches in Plant Biotechnology.** Wiley, Scrivener Publishing Group.pp:185 -216.

11 Satish L, **Ramesh M** (September 01, 2017) Algae based extracts as a natural biostimulant for plant growth and development: Current and future prospects. In: **Photobioreactors:** Advancements, Applications and Research.ISBN:978-1-53612-354-8. Nova Science Publishers, New York, USA. (Eds., Yiu FaiTsang).

12 Satish L and **Ramesh M** (July 15, 2017). Potential of Marine Algae Derived Extracts as a Natural Biostimulant to Enhance Plant Growth and Crop Productivity, p200-211. In

Biotechnology for Sustainability, Achievements, Challenges and Perspectives. Edited by S. Bhore, K. Marimuthu and M. Ravichandran, Published by AIMST University, ISBN: 978-967-14475- 3-6.

13 Shilpha J, Satish L and **Ramesh M** (2017) "Recent Advancements in the Clinical Evaluation of Plant-Derived Anticancer Compounds. In: **Anticancer Plants Clinical Trials and Nanotechnology**, Vol: 3, p 232- 252. ISBN 978-981-10-8215-3. Edited by M S Akthar and M K Swamy. Springer Nature Singapore Pte Ltd.), DOI: 10.1007/978-981-10-8216-0_8.

Faculty Profile as of 01 June, 2020